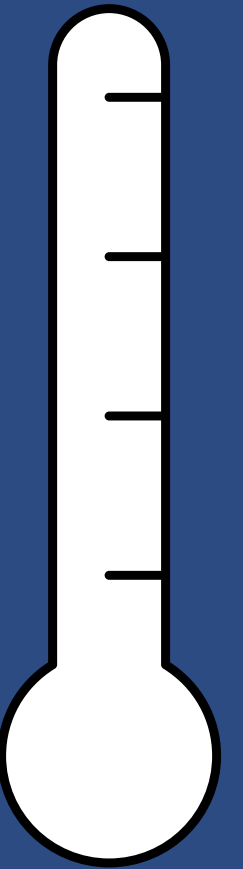


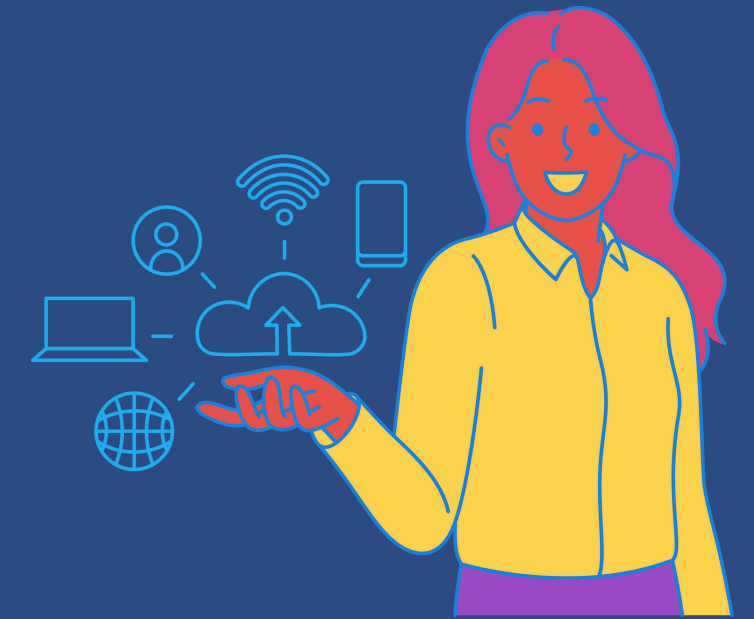


CONSTRUCTION OF ESP32-BASED TEMPERATURE SENSOR

BY ANISHA MARIA LOBO (2120302)

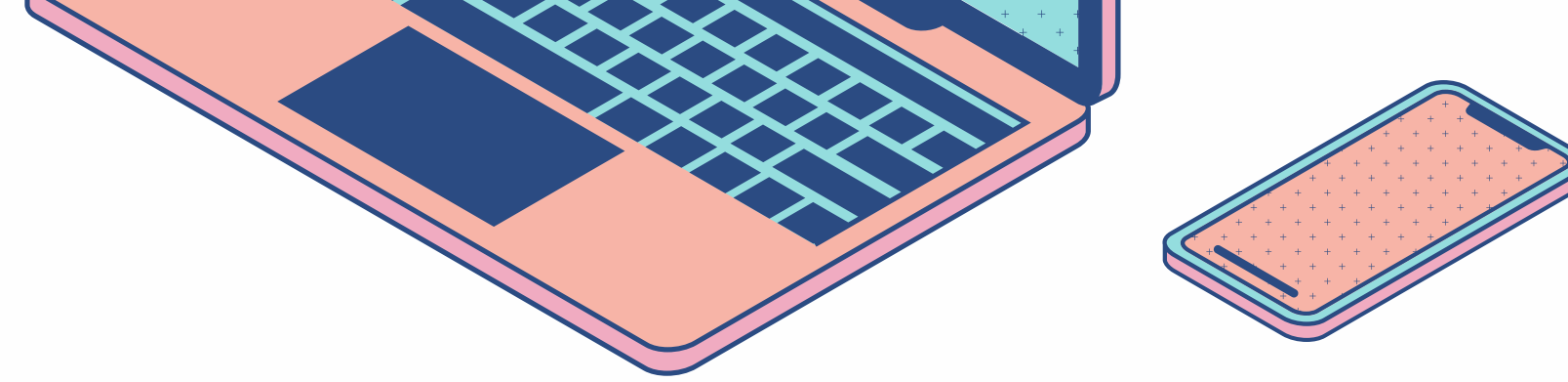


INTRODUCTION



Monitoring environmental parameters such as temperature and atmospheric pressure is crucial for a wide range of applications, including weather forecasting, industrial automation, and indoor climate control.

In this project, we leverage the ESP32 microcontroller along with the BMP280 sensor to create a wireless environmental monitoring system capable of measuring temperature and atmospheric pressure with high accuracy.

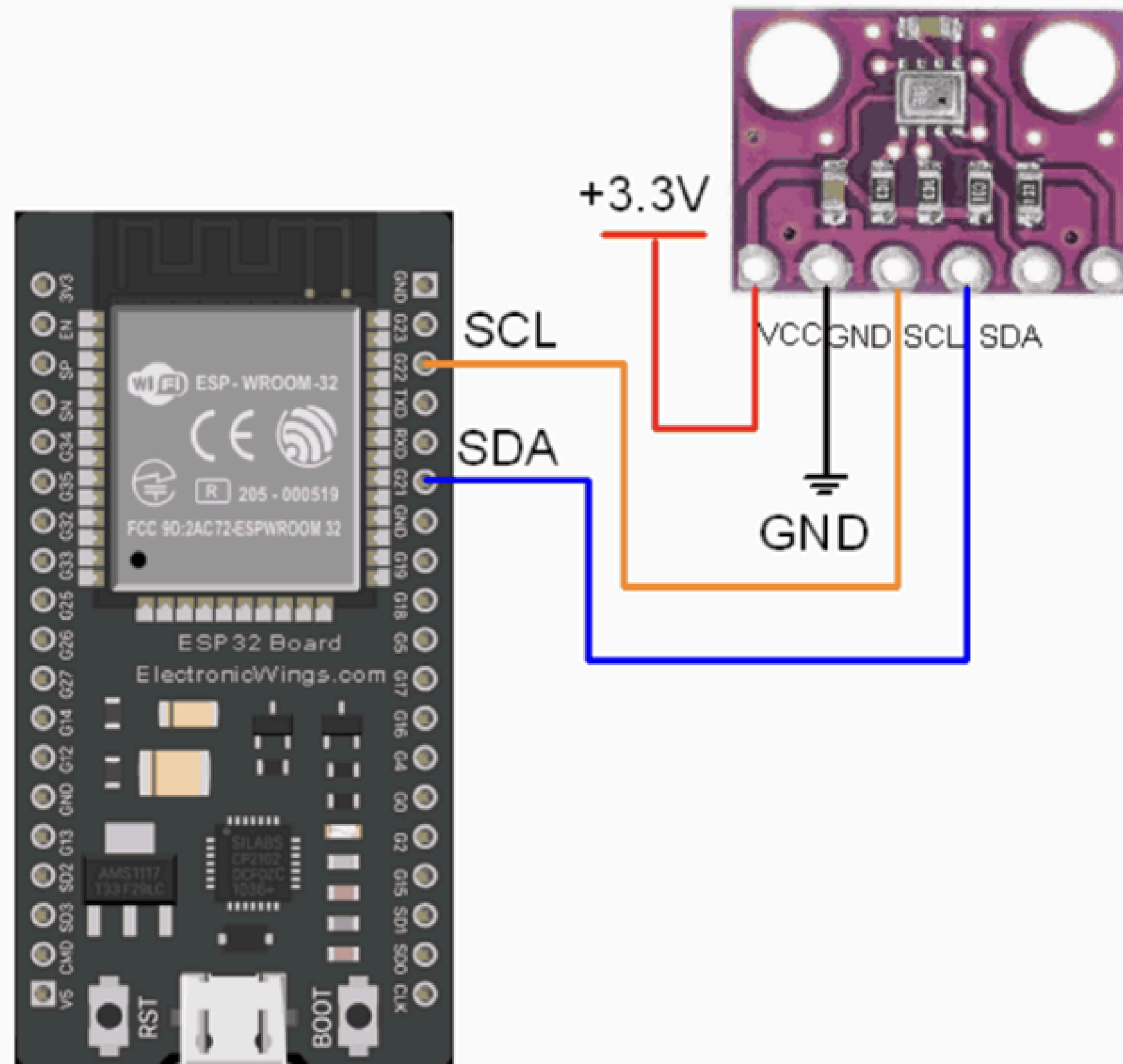


ESP WROOM 32

- Wireless Connectivity
- Security
- Environmental conditions((-40°C to 85°C)
- Flexible Power Management

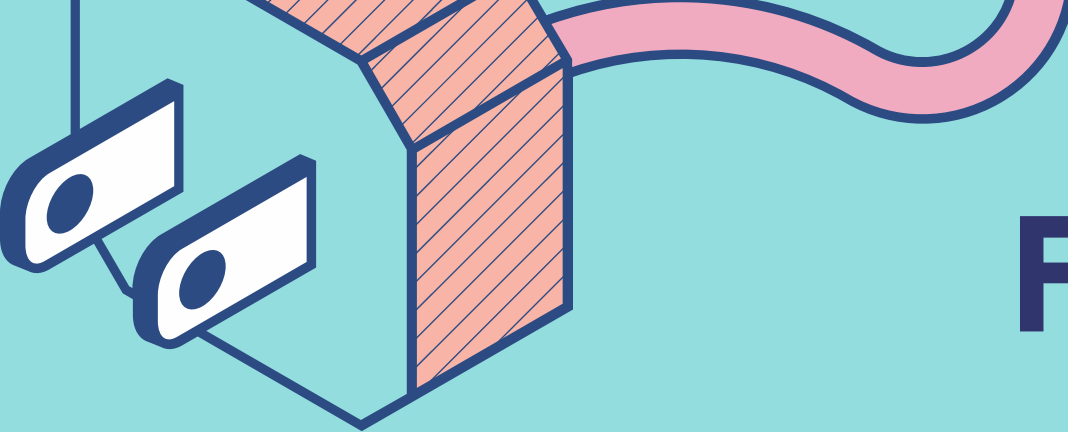
BMP280 SENSOR

- High Accuracy
- Long-Term Stability
- Weather Monitoring
- Data Logging and Analysis



Experiment





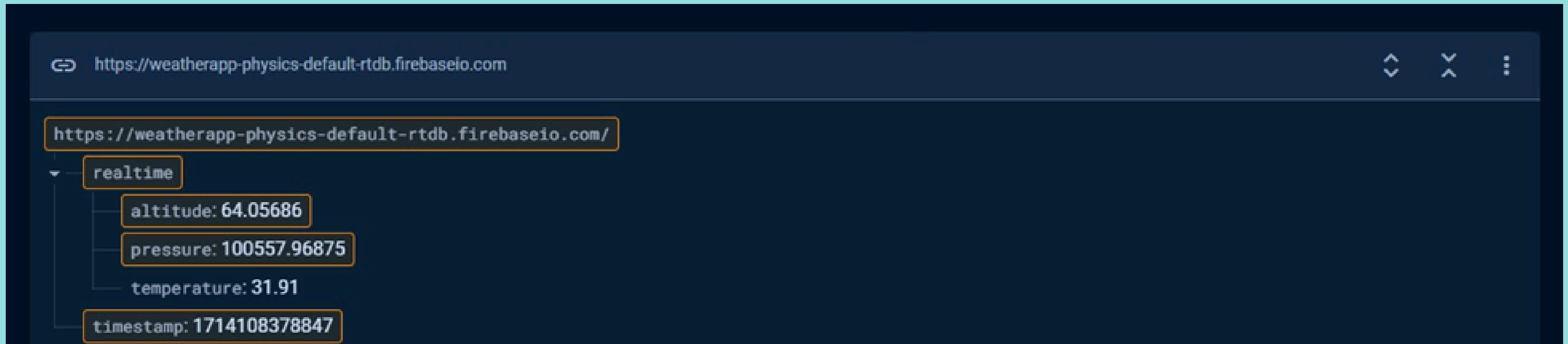
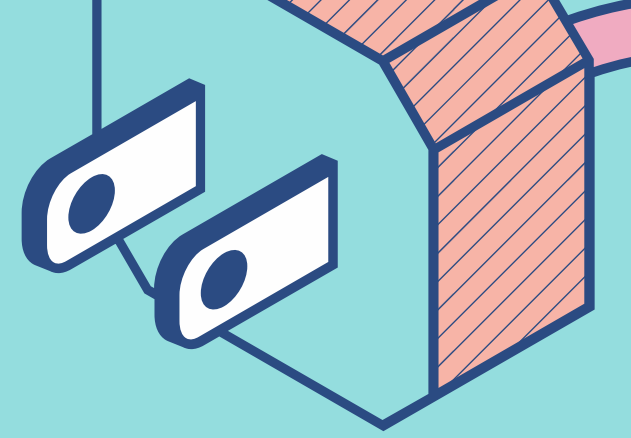
Firestore Data Base



 > environmentDat... > b1713981664  More in Google Cloud 		
 (default)  environmentData  	 b1713981664  	 b1713981664 
+ Start collection	+ Add document	+ Start collection
environmentData >	b1713981664 >	+ Add field
	b1713981977 b1713983426 b1713983433 b1713983443 b1713983458 b1713983457 b1713983466 b1713983473	avgAltitude: 70.02187 avgPressure: 100486.78906 avgTemperature: 32.26913 timestamp: April 24, 2024 at 11:31:04 PM UTC+5:30



Real time Data Base



Weather Application

Filter Date and Time

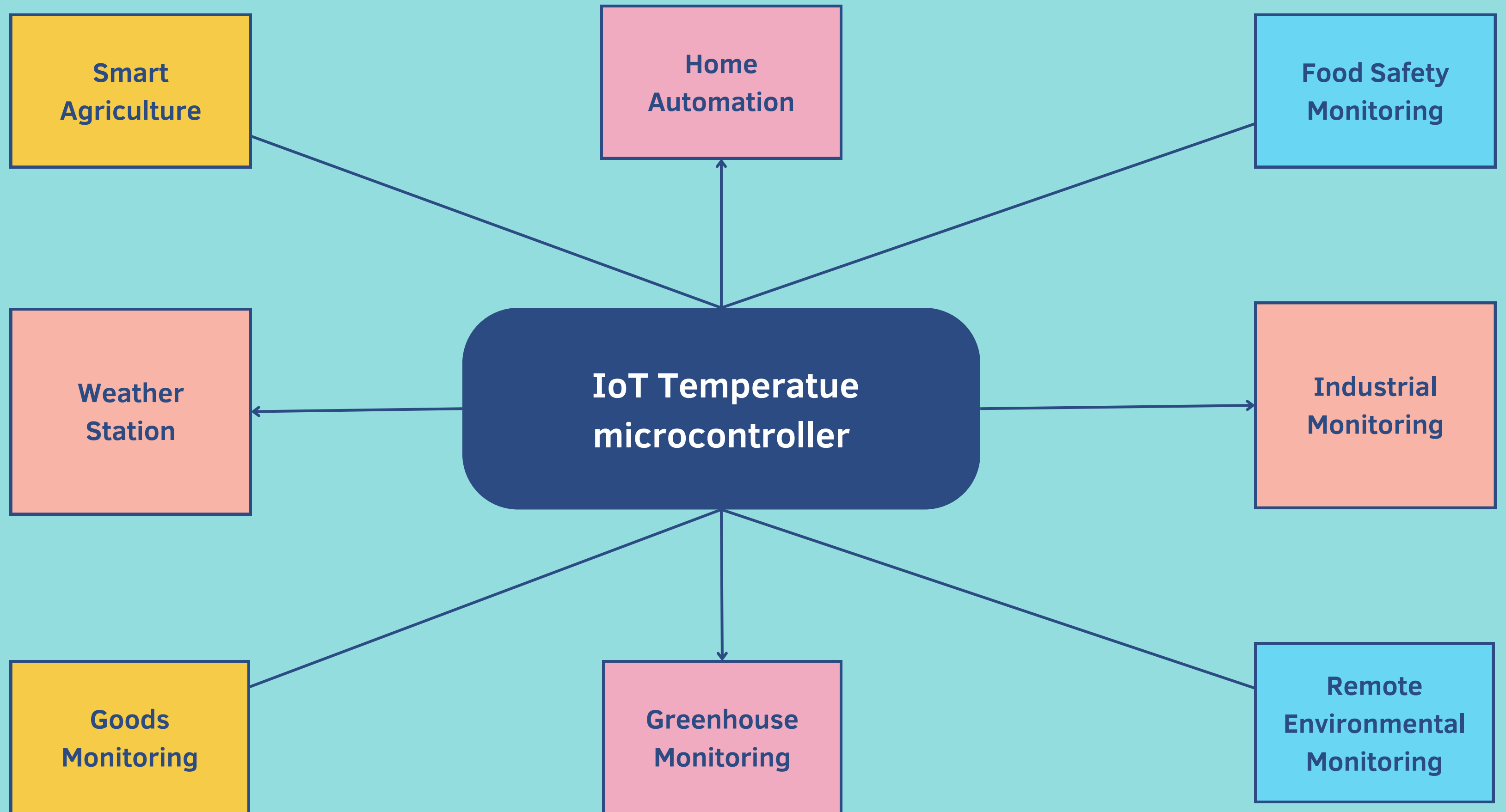
27-04-2024 07:00



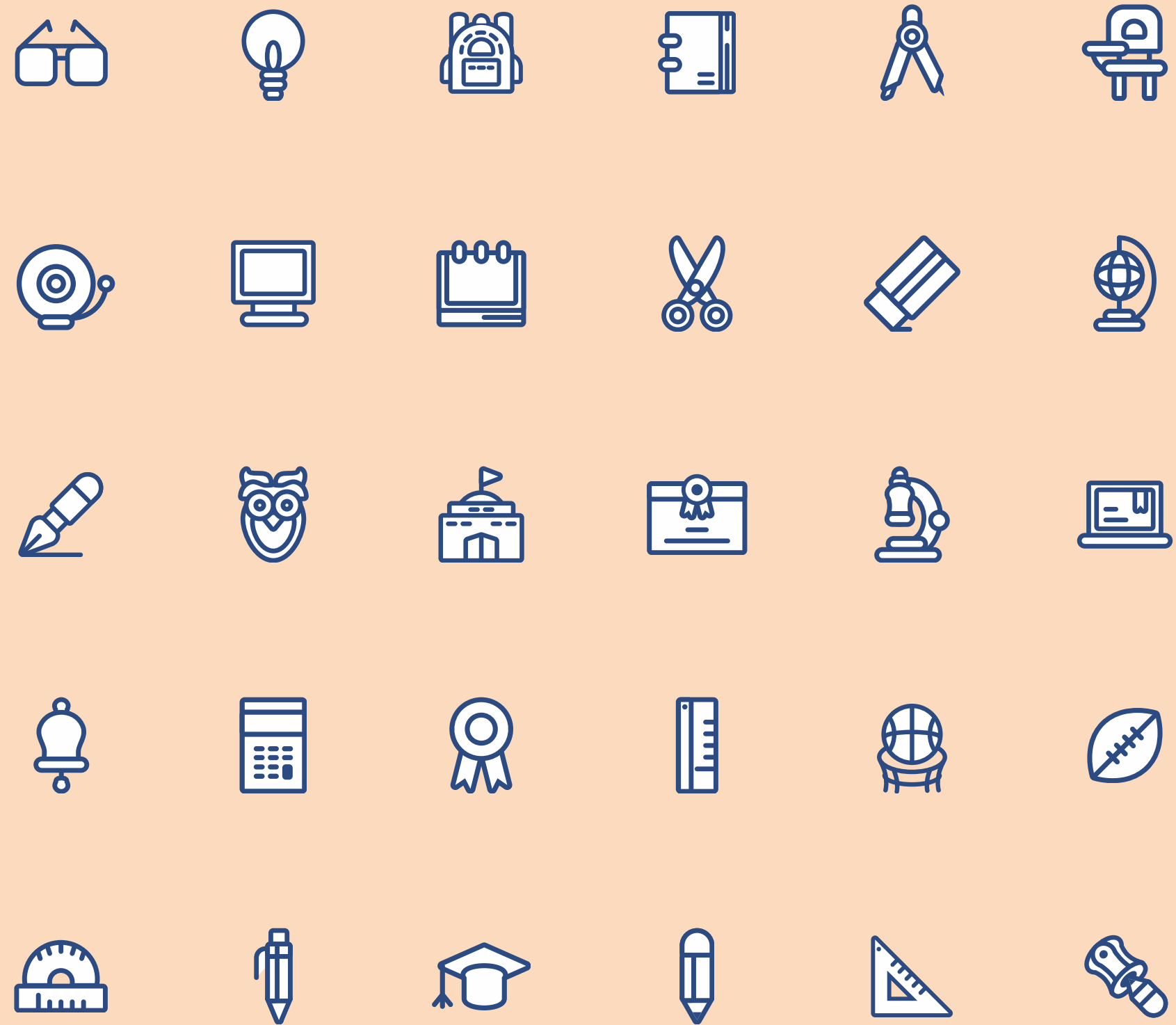
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27/04/2024 07:50	31.03535	100451.10938	73.01327	
27/04/2024 07:55	31.10695	100456.76562	72.5376	
27/04/2024 08:00	31.1855	100459.83594	72.28119	
27/04/2024 08:05	31.17448	100467.47656	71.64095	
27/04/2024 08:10	31.08069	100480.24219	70.57144	
27/04/2024 08:15	31.03703	100489.6875	69.77969	
27/04/2024 08:20	31.0363	100494.96875	69.3355	
27/04/2024 08:25	30.94221	100501.01562	68.83052	

Practical Applications





Other various fields



CONCLUSION

- The integration of the ESP32 microcontroller with the BMP280 sensor harnesses the combined capabilities of both devices, enabling seamless communication and precise data acquisition.
- With real-time temperature monitoring and automated control capabilities, the IoT device enables enhanced efficiency, energy savings, and preventive maintenance
- This device enhance efficiency, energy savings, and preventive maintenance through real-time monitoring and automated control features
- Applications for ESP32-based temperature sensing devices span across industries such as home automation, agriculture, healthcare, and industrial monitoring.
- Remote monitoring capabilities allow users to receive alerts for temperature anomalies or critical events, enabling timely intervention.



Thank You

